



Lab Sheet 4

SQL Operators

Arithmetic Operators

1. Perform addition of two numbers using +.

```
SELECT 10 + 20 AS Addition;
```

2. Perform subtraction using -.

```
SELECT 20 - 10 AS Subtraction;
```

3. Perform multiplication using *.

```
SELECT 10 * 5 AS Multiplication;
```

4. Perform division using /.

```
SELECT 20 / 5 AS Division;
```

5. Calculate the total amount as Price × Quantity.

```
SELECT Price, Quantity, Price * Quantity AS TotalAmount  
FROM Products;
```

Logical Operators

6. Demonstrate AND with two conditions.

```
SELECT *  
FROM Students  
WHERE Course = 'BCA' AND City = 'Delhi';
```

7. Demonstrate OR with two conditions.

```
SELECT *  
FROM Employees  
WHERE Department = 'IT' OR Department = 'HR';
```

8. Demonstrate NOT with a condition.

```
SELECT *  
FROM Students
```

```
WHERE NOT City = 'Delhi';
```

9. Find students belonging to a specific course AND city.

```
SELECT *  
FROM Students  
WHERE Course = 'BCA' AND City = 'Delhi';
```

10. Find employees belonging to either of two departments.

```
SELECT *  
FROM Employees  
WHERE Department = 'IT' OR Department = 'HR';
```

Comparison Operators

11. Demonstrate the = operator.

```
SELECT *  
FROM Students  
WHERE Course = 'BCA';
```

12. Demonstrate the > operator.

```
SELECT *  
FROM Employees  
WHERE Salary > 50000;
```

13. Demonstrate the < operator.

```
SELECT *  
FROM Employees  
WHERE Salary < 50000;
```

14. Demonstrate >= and <=.

```
SELECT *  
FROM Employees  
WHERE Salary >= 30000 AND Salary <= 60000;
```

15. Demonstrate the <>/!= operator.

```
SELECT *  
FROM Students  
WHERE City <> 'Delhi';
```

Special Operators

16. Demonstrate BETWEEN for salary range.

```
SELECT *
FROM Employees
WHERE Salary BETWEEN 30000 AND 60000;
```

17. Demonstrate IN for selecting multiple cities.

```
SELECT *
FROM Students
WHERE City IN ('Delhi', 'Mumbai', 'Pune');
```

18. Demonstrate NOT IN.

```
SELECT *
FROM Students
WHERE City NOT IN ('Delhi', 'Mumbai');
```

19. Demonstrate LIKE for names beginning with a particular letter.

```
SELECT *
FROM Students
WHERE Name LIKE 'A%';
```

20. Demonstrate IS NULL and IS NOT NULL.

IS NULL:

```
SELECT *
FROM Employees
WHERE ManagerID IS NULL;
```

IS NOT NULL:

```
SELECT *
FROM Employees
WHERE ManagerID IS NOT NULL;
```

Set Operations

21. Demonstrate UNION using two compatible queries.

```
SELECT City
FROM Students
UNION
SELECT City
FROM Employees;
```

22. Demonstrate UNION ALL.

```
SELECT City
FROM Students
UNION ALL
SELECT City
FROM Employees;
```

23. Demonstrate INTERSECT where supported by the DBMS.

```
SELECT City
FROM Students
INTERSECT
SELECT City
FROM Employees;
```

24. Demonstrate EXCEPT/MINUS where supported.

EXCEPT:

```
SELECT City
FROM Students
EXCEPT
SELECT City
FROM Employees;
```

For MySQL:

```
SELECT DISTINCT City
FROM Students
WHERE City NOT IN (
    SELECT City
    FROM Employees
);
```

Problem using Multiple Operators

Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operation.

```
SELECT Name, Department, Salary,
       Salary * 1.10 AS SalaryWithBonus
FROM Employees
WHERE (Department = 'IT' OR Department = 'HR')
AND Salary BETWEEN 30000 AND 80000
AND Name LIKE 'A%';
```

OUTPUT:-

Arithmetic Operators

1. Perform addition of two numbers using +.

```
Addition
-----
30
```

2. Perform subtraction using -.

Subtraction

10

3. Perform multiplication using *.

Multiplication

50

4. Perform division using /.

Division

4

5. Calculate the total amount as Price × Quantity.

Price	Quantity	TotalAmount
-----	-----	-----
35.50	2	71.00
40.50	1	40.50
50.00	3	150.00

Logical Operators

6. Demonstrate AND with two conditions.

StudentID	Name	Course	City
-----	----	-----	----
1	Amit	BCA	Delhi

7. Demonstrate OR with two conditions.

EmployeeID	Name	Department
-----	----	-----
1	Rahul	IT
2	Priya	HR

8. Demonstrate NOT with a condition.

StudentID	Name	Course	City
-----	----	-----	----
2	Priya	BCA	Mumbai
3	Rahul	BBA	Pune

9. Find students belonging to a specific course AND city.

StudentID	Name	Course	City
-----	----	-----	----
1	Amit	BCA	Delhi

10. Find employees belonging to either of two departments.

EmployeeID	Name	Department
-----	----	-----
1	Rahul	IT
2	Priya	HR

Comparison Operators

11. Demonstrate the = operator.

StudentID	Name	Course
-----	----	-----
1	Amit	BCA
2	Priya	BCA

12. Demonstrate the > operator.

EmployeeID	Name	Salary
-----	----	-----
1	Rahul	60000
3	Amit	75000

13. Demonstrate the < operator.

EmployeeID	Name	Salary
-----	----	-----
2	Priya	40000
4	Neha	35000

14. Demonstrate >= and <=.

EmployeeID	Name	Salary
-----	----	-----
1	Rahul	50000
2	Priya	40000
3	Amit	60000

15. Demonstrate the <>/!= operator.

StudentID	Name	Course	City
-----	----	-----	----
2	Priya	BCA	Mumbai
3	Rahul	BBA	Pune

Special Operators

16. Demonstrate BETWEEN for salary range.

EmployeeID	Name	Salary
-----	----	-----

1	Rahul	50000
2	Priya	40000
3	Amit	60000

17. Demonstrate IN for selecting multiple cities.

StudentID	Name	City
-----	----	----
1	Amit	Delhi
2	Priya	Mumbai
3	Rahul	Pune

18. Demonstrate NOT IN.

StudentID	Name	City
-----	----	----
3	Rahul	Pune
4	Neha	Jaipur

19. Demonstrate LIKE for names beginning with a particular letter.

StudentID	Name
-----	----
1	Amit
5	Anjali

20. Demonstrate IS NULL and IS NOT NULL.

IS NULL:

EmployeeID	Name	ManagerID
-----	----	-----
4	Neha	NULL

IS NOT NULL:

EmployeeID	Name	ManagerID
-----	----	-----
1	Rahul	101
2	Priya	102
3	Amit	101

Set Operations

21. Demonstrate UNION using two compatible queries.

City

Delhi
Mumbai
Pune
Jaipur

22. Demonstrate UNION ALL.

```
City
----
Delhi
Mumbai
Pune
Delhi
Mumbai
```

23. Demonstrate INTERSECT where supported by the DBMS.

```
City
----
Delhi
Mumbai
```

24. Demonstrate EXCEPT/MINUS where supported.

```
City
----
Pune
Jaipur
```

Problem using Multiple Operators

Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operation.

Name	Department	Salary	SalaryWithBonus
----	-----	-----	-----
Amit	IT	50000	55000
Anjali	HR	60000	66000